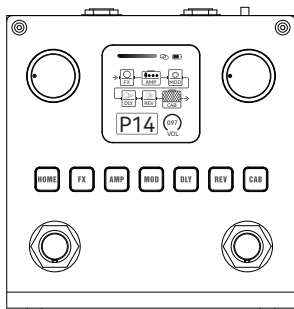


M-VAVE

USER MANUAL

ANNBLACK BOX

Guitar/Bass AMP Modeler Multi Effects Processor



Website: www.m-vave.com

M-EFCS for Windows/Mac OS: <http://www.m-vave.com/app> download

M-EFCS for Android/iOS: scan the QR code



ANNBLACK BOX is a portable and multifunctional Guitar/Bass effector. It has 80 editable Presets, the users also can customize effectors chain and save. It has built-in battery, it can last up to 10 hours on a full charge. ANNBLACK BOX has 6 effects Modules, it provides the Noise Gate, Boost, Compressor, 20 Preamp, 3-band EQ, 7 Modulations, 5 Delays, 6 Reverbs, 20 IR CAB slots and so on.

Press the HOME button on the Main Display Screen to enter the Module Operation Screen, then use the PARA Knob to customize the position of the effects Module. At this time, you can also select any Module by pressing the Right-Footswitch and then open/close this module with the Left-Footswitch, which is convenient for the user to control the Module on/off with the foot during the performance(Stomp Mode). When you click the MOD/DLY Button, the MOD/DLY Module opens on the screen, you can tap the Right-Footswitch to adjust the effect speed of the MOD/DLY Module.

You can download the computer software or mobile APP from our official channels, use it to edit presets, swap presets, share presets, import/export presets, and import the IR files, import the AMP module timbre data(The file format is AM2,AM3,AMx,etc), besides you can restore the factory Settings or firmware upgrade on the computer. It support BT wireless music, earphone monitoring function, and computer sound card, etc. ANNBLACK BOX can be used as an audio interface with a cellphone, use a OTG converter (separately buying) and accessory USB cable to connect the cellphone, then enter any video/audio recording APP for direct recording. It integrates a variety of function, it is easy-carry and it has excellent sound! It could be a good partner of your music instruments.

EN Attentions

■ The power supply:

It is equipped with built-in battery for power supply, which can last for 10 hours after full charge. It can charge with the 5V adapter or the power bank. When the battery run low, you can use the 5V 2A adapter, you need 5 hours to fully charge the battery. When the battery run low, don't use it for a long time to avoid unstable working performance, please charge it in time.

■ The connection:

Be sure to turn off the power of related equipment before connecting or disconnecting the device.

■ The electrical interference:

The electromagnetic wave of radio, television and other equipment may cause interference to the equipment, so please try to stay away from radio, television and other equipment.

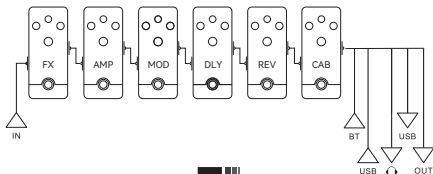
■ The location:

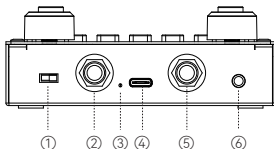
To avoid deformation, discoloration, or other serious damage, do not expose this unit to the following conditions:

- a. Heat sources b. Magnetic fields c. Sunlight
- d. High temperature or moist district
- e. Excessively dusty or dirty locations
- f. Strong vibrations or shocks

EN Effectors chain diagram

(The order of the 6 effects Modules can be customized.)





① **Power switch.**

② **IN:** 1/4" MONO audio input jack.

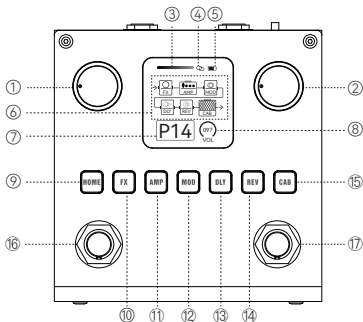
③ **Charging indicator:** When the battery is normally using, the indicator will go off; During charging, the indicator will stay red, then it will change to a different color after fully charging.

④ **USB jack:** Use to charge the battery/The ANNBLACK BOX can be sound card of computer/Computer software can be connected for interactive data transmission/OTG recording/support MIDI Pedal.

Tips: ANNBLACK BOX can be used as an audio interface with a cellphone, use a OTG converter (separately buying) and accessory USB cable to connect the cellphone, then enter any video/audio recording APP for direct recording or playback music.

⑤ **OUT:** 1/4" MONO audio output jack.

⑥ **Earphone output.**



① **Master**, Control the volume. (Note: The BT volume and USB input volume are not controlled by this knob.)

② **PARA knob (Rotary encoder with button)**, Rotate it to select items or adjust parameters, Click it to enter, exit, or confirm the items, Press and hold for 1 second to display the "Save to" screen, Press and hold for 2 second to display the "Swap to" screen. When on the "Save to" or "Swap to" screen, if you click the HOME button, you can exit and return to "Main Display Screen".

Save to

<P08>

P08 Swap to

<P09>

③ **IN jack input volume real-time display bar.**

④ **Wireless BT pattern**, When the pattern is off, it indicates that BT is not enabled. A white pattern indicates that BT has started but is not connected. The blue pattern indicates that BT is successfully connected.

⑤ **Battery status pattern.**

⑥ **This diagram shows the order and the on/off state of the 6 effect Modules.** If the pattern is off, the effect is not enabled. If the pattern is on, the effect is started. The state of the pattern corresponds to the state of the 6 buttons down on the right. If all 6 patterns are Off, the ANNBLACK BOX is in Bypass state.

⑦ **Preset number,** When this number is selected using the PARA knob, rotate it to quickly switch Presets. ANNBLACK BOX includes 80 Presets, with the first 40 (P01~P40) containing default factory effects, thereinto the P36~P40 were designed for bass. When you need the factory reset, you can separately reset these 40 Presets to the default values. If you want reset all the 80 Presets, it is also feasible.

Note: Enter the mobile APP or computer software interface to perform this operation.

⑧ **The volume of current Preset.**

⑨ **HOME illuminated Button / BT switch Button(with BT status light),** click the HOME button to display the Main Display Screen on the screen, then press the HOME button again to enter the Module Operation Screen. Besides, you can hold this button to open/close the BT. When the button light off, the BT had been turned off; when the button light flashing, it means BT is turning on, but not in connection; when the button light keep on, it means BT has been connected.

"Main Display Screen" is the home page displayed right after the power is turned on. You can directly see the key information and settings of the current Preset, you can also know about battery conditions and other information and so on.



Module Operation Screen: In this screen, you can easily operate the connection order of 6 modules, you can also open/close a module freely by using two Footswitches (Stomp Mode).



⑩ FX Module illuminated Button, Click to enter the Edit Settings Menu of Module, then click again to turn on/off Module. At any time, you can hold the button to quickly turn on/off Module. If this Button is lighting on, it means the Module is turned on. If the light goes off, the Module is turned off. When the Edit Settings Menu is displayed on the screen, you can use the PARA encoder to adjust any of the effects and parameters on the screen. At this time, you can also use the Left Footswitch to turn the current Module on/off.

Tips: The buttons of the 6 effects Modules all support the above functions.

⑪ AMP Module illuminated Button.

⑫ Modulation Module illuminated Button, When the Screen displays the MOD's Edit Settings Menu, you use the Right Foot switch, you can also Tap to obtain the Speed parameter of the effect. At this time, if the Module has been turned on, the button light will also blink based on the Speed parameter.

⑬ Delay Module illuminated Button, When the Screen displays the DLY's Edit Settings Menu, you use the Right Foot switch, you can also Tap to obtain the Time parameter of the effect. At this time, if the Module has been turned on, the button light will also blink based on the Time parameter. When the Screen displays the DLY's Edit Settings Menu and the DLY had been turned on, you can hold the Right-footswitch to start the Hold function, after a long step of 1.5 seconds the machine will read the Delay wet sound at this moment. If you don't release the foot, you'll keep this Hold sound. If you release the foot, you can immediately cancel the Hold sound(Freeze function).

⑭ REV Module illuminated Button.

⑮ IR CAB Module illuminated Button.

⑯ Left Footswitch, it use to turn down the Presets when in Main Display Screen. In some other interfaces, it can be used for on/off Modules. You can press the left and right Footswitches at the same time to enter Tuning Mode, and then click either Footswitch to exit tuning.

⑦ **Right Footswitch**, it use to turn up the Presets when in Main Display Screen. When in the Module Operation Screens, it can be used to select Modules. When in the MOD/DLY Module screen, you can tap the Right-Footswitch to adjust the effect speed of the MOD/DLY Module. When in the DLY Module Screen and the DLY is turned on, you can hold the Right-footswitch to enable the Hold function (freezing the wet sound to get the unlimited voice).

Note: On some interfaces, the Right-Footswitch does not work.

Tips: The tuning interface is shown in the picture.

There are two items to choose from in this interface, the "refer to" item is to select the standard A's frequency 430-450Hz, the "Mode" item is to select the voice output status Bypass or Mute.



EN Effectors module

I: FX Module(FX module can select Noise Gate or NG+Boost or NG+Compress, three collocation ways,you can use 6+1 effect at the same time.)

1. Noise Gate++

- Gate parameter, Adjust the intensity of AI Noise Gate, When set to 0, the effect is turned off.

2. Boost(Noise Gate can be added)

- Gate parameter, Adjust the intensity of AI Noise Gate, When set to 0, the Noise Gate is turned off.
- Gain parameter, Adjust the gain of the Boost.

3. Compress(Noise Gate can be added)

- Gate parameter, Adjust the intensity of AI Noise Gate, When set to 0, the Noise Gate is turned off.
- Sustain parameter, Adjust the compression amount.
- Attack parameter, The starting time of the compressor after the signal exceeds the threshold, the larger the value, the slower the compression starting time, and the more prominent the auditory sound-head; The smaller the value, the faster the startup time.

- Level parameter, Use this parameter to adjust the output volume of this Module again.

4.AI Gate MS, the intelligent algorithm based on the statistical deviation tracking model.

● Gate: The threshold for opening the Gate. It continuously monitors the level intensity of the input signal. The higher the threshold is set, the more stringent the conditions for the Gate to open, and only stronger signals can pass through. Conversely, the lower the threshold, the more signals can pass.

● Bias: By setting the Bias lead, the AI noise reduction model can be made to "prepare" in advance for the upcoming noise or signal changes. In audio noise reduction, if the changing trend of the noise can be predicted in advance and the bias is adjusted accordingly, the model can more accurately suppress the noise when it appears, reducing the interference of the noise on the useful signal and improving the timeliness and accuracy of noise reduction.

II: AMP Module (1-15 is guitar preamp, 16-20 for bass preamp)

- Gain
- Level
- Bass
- Mid
- Treble

III: Modulation Module(Support TAP function)

1. Chorus

- Speed
- Depth
- Mix: Adjust the dry and wet ratio.

2. Phaser

- Speed
- ParamA: MidCut, the larger the value, the greater the

midfrequency reduction of the effect, this parameter is mainly designed for distortion timbre.

- ParamB: Resonance, resonance intensity adjustment of phase effect.
- Feedback: Adjust the feedback amount of Phaser.

3. Tremolo, it is basically a change of the signal level controlled by an Low-frequency Oscillator.

- Speed
- Depth
- Level parameter, Use this parameter to adjust the output volume of this Module again.

4. Flanger, it is use the same principle as Chorus, but it uses a shorter delay time and adds regeneration(or repeats) to the modulating delay. This results in an exaggerated up and down sweeping motion to the effect.

- Speed
- Depth
- Feedback: Adjust the amount of feedback for the sweep effect in wet sound.
- Mix: Adjust the dry and wet ratio.

5. Vibrato, The vibrato effect modulates the pitch of the incoming signal. The result is similar to the Tremolo technique used by vocalists. In contrast to a Chorus or Flanger effect, no direct signal is combined with the pitch-modulated signal.

- Speed
- Depth

6. Univibe, a unique timbre, a combination of chorus and phaser.

- Speed
- Depth
- Mix

7. Autofilter, filter works regularly by setting the frequency, both guitar and bass can be used.

- Speed

- **Min:** Minimum frequency of the filter sweep frequency, when the value is turned up, the middle and low frequency parts of the wet sound are more prominent.
- **Max:** Maximum frequency of the filter sweep frequency, when the value is turned up, the middle and high frequency parts of the wet sound are more prominent.
- **Mix**
- **Feedback**

IV: Delay Module(Support TAP and Freeze function)

1. Analog, it is the delay effect of the signal of the analog tube, and the timbre has the characteristics of retro and warm.

- **Time,** adjust the feedback speed of echo repeats, 600–60bpm.
- **Feedback,** the feedback times of echo repeats.
- **Mix,** Adjust the dry and wet ratio.
- **Phaser,** the Phaser effect is increased in the Delay wet sound. This parameter can adjust the amount of Phaser used.
- **Pitch:** We add a little detuning effect to the delay wet sound, this parameter adjusting the pitch of the detuning effect, which is a slightly shifted sound, a retro and charming sound.

2. Ducking Delay, When we process the delayed wet sound, we add the Noise Gate to the part before the wet sound. Thus, the front part of the wet sound is suppressed to achieve a front evasive effect, and the sound is gradually increased, the dynamics of the delay are more ‘felt’ than ‘heard’.

- **Time,** adjust the feedback speed of echo repeats, 600–60bpm.
- **Feedback,** the feedback times of echo repeats.
- **Mix,** Adjust the dry and wet ratio.
- **Unpack,** After this parameter is turned up, the sudden rise in the volume of the wet rear part can be unpacked, so as to achieve a smoother delay dynamic.
- **Speed,** the Chorus effect is increased in the Delay wet sound. This parameter can adjust the speed of Chorus used.
- **Depth,** this parameter adjusts the depth of the Chorus.

3. dTape, Tape delay, it replicates the unique delay style of tape machine, you can not only get the warmth and silky of

professional grade tape machine, but also simulate the sound effects of tape in real situations such as CRINKLE, BIAS, and so on.

- Time, adjust the feedback speed of echo repeats, 600–60bpm.
- Feedback, the feedback times of echo repeats.
- Mix, Adjust the dry and wet ratio.
- Grit, the GRIT takes on the function of TAPE BIAS. This parameter adjusts tape machine bias, from under biased to over-biased, it sounds like an overdrive of wet sounds. Bias sets the dynamic range and headroom of the wet signal.
- Speed, the SPEED takes on the function of TAPE CRINKLE. This parameter adjusts the amount and severity of tape irregularities, including friction, creases, splices and contaminants. Tape Crinkle characteristics track accordingly to tape speed. Set to minimum for a fresh, clean tape. Set to maximum for a tape that has been mangled and chewed for years.
- Depth, This parameter control varies the amount of mechanically related tape speed fluctuations. This also results in natural tape machine style modulation, it's like a chorus. Parameter minimum for a more tuned, cleaned and serviced tape machine. Parameter maximum to hear the sound of a tape machine in need of service. In between the extreme settings, a natural tape modulation is achieved.

4. Dual Delay, two independent delay echoes, the time-lag of the first echo and the second echo achieves a very interesting delay rhythm effect.

- Time, adjust the feedback speed of echo repeats, 1200–120bpm.
- Feedback, the feedback times of echo repeats.
- Mix, Adjust the dry and wet ratio.
- Time-Mode, Adjust the time-lag between the two echoes, the minimum is the equivalent of no time-lag, and the greater the number, the greater the time-lag.
- Speed, the Chorus effect is increased in the Delay wet sound. This parameter can adjust the speed of Chorus used.
- Depth, this parameter adjusts the depth of the Chorus.

5. Lofi Delay, a special, retro and destructive delay effect, the wet sound reflects the filter, the vinyl record, the noise lo-fi and many

other senses.

- Time, adjust the feedback speed of echo repeats, 600–60bpm.
- Feedback, the feedback times of echo repeats.
- Mix, Adjust the dry and wet ratio.
- Grit, When the parameter is turned up, it sounds like a overdrive wet sound.
- Speed, the Chorus effect is increased in the Delay wet sound. This parameter can adjust the speed of Chorus used.
- Depth, this parameter adjusts the depth of the Chorus.

V: Reverb Module

1. Room reverb, it simulates a relatively small, simple-structured room sound where many reflections are absorbed by soft material in the room, and sound is reflected by walls.

- Decay, Adjust the duration of echo.
- Mix, Adjust the ratio of wet and dry.
- HPass, HIGH pass, only used to regulate wet sound.
- LPass, Low pass, only used to regulate wet sound
- Depth, the pitch of wet sound produces a small cyclical rise and fall, you will hear vintage and charming background sound.

2. Hall reverb, it gives a wide, slightly scattering feeling, it simulates a grand ambient sound.

- Decay
- Mix
- HPass
- LPass
- Depth

3. SWELL reverb, the Swell machine brings in the reverb gradually behind the dry signal for subtle evolving textures, like having a volume pedal on the wet sound.

- Decay
- Mix, If the dry sound is removed, and set the RiseT parameter to 0, it can mimic the sound of string instruments such as the violin.

- HPass
- LPass
- RiseTime, it adjusts the rise time of the swelled signal, I suggest you choose shorter times for single-line soloing or longer times for ambient chord work.

4. Spring reverb, it is a common type of reverberation. The sound signal is transmitted to the spring tank, and the pickup picks up the resonant sound of the spring tank, to mimics the reverberation effect produced in space.

- Decay
- Mix
- HPass
- LPass
- Combs, control the number of springs.

5. Shimmer reverb, there is a pitch-shift sound in the wet sound. Adjust the pitch of the pitch-shift sound, you can get the dissonant interval, create a scary background sound. You also can get the harmonic interval, it's a wonderful sound, a resplendent and unearthly ambience. The Amount parameter allow for a range of shimmer effects from laid-back and subtle to full-blown majestic splendor.

- Decay
- Mix
- Tone, Adjust the brightness of the reverb wet sound.
- Pitch, Adjust the pitch of the pitch-shift sound, the minimum value is the same as the original sound, the maximum value is two octaves above the original sound.
- Amount, adjust the amount of pitch-shift sound.

6. Cloud reverb, is a gorgeous large ambient reverberation, it sounds like the music come from all sides of the cloud. Cloud reverb can take any modest guitar or synth sound and turn it into a gorgeous ensemble.

- Decay
- Mix

- HPass
- LPass
- Diffusion, softens the early reflections to create a more diffused reverb. As Diffusion is increased, the reverb is smoothed and softened, the delay and reverberation mix together more naturally.

VI: IR CAB Module (1-15 is guitar CAB simulator, 16-20 is bass CAB simulator)

- Level
- Low cut, Adjust the low cut point, optionally 20-300 Hz. If you choose 300 Hz, it means that low frequencies below 300 Hz are suppressed.
- High cut, Adjust the high cut point, optionally 18000-5000 Hz. If you choose 5k Hz, it means that high frequencies above 5000 Hz are suppressed.

EN Technical parameters

Input	Standard 1/4" jack MONO/TS
Output	Standard 1/4" jack MONO/TS, 1/8" Stereo jack
Sample rate	44.1KHz/24bit
Size	113(L)x109(W)x41(H)mm
Weight	410g
Power supply& USB data transmit	Type-C、5V≥300mA
Battery capacity	2000mah
Battery Model/Type	103450
Battery Nominal Voltage	3.7V
Battery life	10+ hours
Charging current limit	About 408 mA
Dynamic Range	105dB
Total effects	60+
Effects Slots/Effects Modules	Total of 6+1 simultaneous effects slots, it can customized the order
Presets amount	40+40
USB AUDIO (OTG)	44.1K/24Bit, support Loopback ON/OFF
USB MIDI	Support
BT music	Support
Software support	WINDOWS,MAC,IOS,Android
IR processing	44.1K/24Bit Mono WAV files, 1024 sampling points
Power supply	Built-in battery or USB 5V
AMP modeler	Totally 20 (1-15 is electric guitar preamp, 16-20 for bass preamp)
IR slots	Totally 20 (1-15 is electric guitar CAB, 16-20 for bass CAB)
Working current	About 200 mA
Attachment	USB cableX1

● APP using tips:

Before operation, make sure that the BT of the cellphone and ANNBLACK BOX are both turned on, then run the APP and click the “≡” button, click the "Current Device" and select the "ANNBLACK BOX", then scan nearby devices, you can connect to use.



scan the QR code

● Can support wired/wireless connection MIDI pedal (MIDI pedals need to be purchased separately, Chocolate Plus pedal as an example), so as to achieve the purpose of expanding the Footswitches.

1. The following functions can be achieved: 4 Presets can be realized as a Bank, Bank+ and Bank-, all the Presets can be freely loaded.

2. Six modules can be switched on and off by pressing the external footswitches.

3. You can use the first footswitch to replace the Home button, the second Footswitch to replace the PARA button "press" action, and the third and fourth to replace the PARA button "turn counterclockwise" and "turn clockwise".

● **Tips:** Loopback function, a common function on sound cards, can be enabled or disabled in software, it is set to the "on" position in factory condition.

When the function is turned on, the music in the computer/mobile phone will be recorded at the same time with the instrument sound, it's the same track mixed recording, including USB and BT music.

If the function is turned off, when using the Blackbox as a sound card for recording, only the IN-interface instrument sounds can be recorded, which is more convenient for sub-track recording of instrument sounds during computer recording.

● We will continue to improve our products and reserve the right to modify certain functions and specifications (including but not limited to appearance, packaging design, operation manual,

accessories, dimensions, specifications, displays, etc.) without prior notice. Please confirm the product functions and specifications with the local distributor before the purchase. Product pictures and color due to the light error and screen admiration setting. It may be different from the physical product effect. Please refer to the physical object accordingly.

EN FCC Warning Statement

★ Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment. This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

FCC Radiation Exposure Statement

The device has been evaluated to meet general RF exposure requirement. The device can be used in portable exposure condition without restriction.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

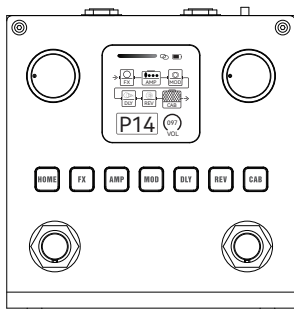
- (1) This device may not cause harmful interference.
- (2) This device must accept any interference received, including interference that may cause undesired operation.

M-VAVE

USER MANUAL

ANNBLACK BOX

数字建模多功能吉他/贝斯效果器说明书



Website: www.m-vave.com

M-EFCS for Windows/Mac OS: <http://www.m-vave.com/app> download

M-EFCS for Android/iOS: scan the QR code



ANNBLACK BOX是一款多功能吉他/贝斯效果器。ANNBLACK BOX有80个出厂预设，用户可自定义音色进行覆盖储存，也可以对预设音色进行编辑和保存。本产品内置蓄电池，充满电后可持续使用续航10小时。ANNBLACK BOX具备6个效果模块，提供了降噪效果器、Boost效果器、压缩效果器、20种经典的箱头音色模拟、三段EQ调节器、7种调制效果、5种延时效果、6种混响、20种经典的IR箱体模拟等效果音色。

在主界面时，再次短按HOME键后，进入模块操作界面，可用PARA旋钮自定义移动效果模块的位置，此时也可用右踩钉循环选择任一模块然后用左踩钉开/关此模块，方便用户在演出时用脚控制模块的开/关（Stomp Mode 单块模式）。当您短按MOD/DLY按键后，MOD/DLY模块在屏幕上打开时，可Tap右踩钉打点获取MOD/DLY模块的效果速度。

可从我司官方渠道免费下载配套的电脑软件或者手机APP软件，支持使用软件编辑、交换、共享、导入/导出音色数据、导入IR文件和导入AMP模块音色数据（AM2,AM3,AMx音色文件），使用电脑软件还可对本机恢复出厂设置和进行固件升级等等。另外它还带有无线BT音乐播放功能，耳机监听功能，手机OTG内录功能，电脑声卡功能等等。多种功能效果集于一身，体积小性能好，便于携带且音色优秀，是您不错的乐器伴侣。

CN 注意事项

■ 电源供应:

本机设有内置电池供电, 充满电后可持续使用10小时, 建议使用5V 2A充电器或充电宝为本机进行供电, 本机支持充电&用电同步进行。在本机内置电池耗尽的状态下进行充电, 大概5小时可以充满。电量不足时, 请勿长时间使用, 以避免不稳定的工作性能, 请及时充电。

■ 连接:

当连接或断开设备前, 请先关闭其他相关设备。

■ 电器干扰:

收音机、电视机等电磁波可能会对设备使用造成干扰, 因此请尽量远离收音机和电视机等设备。

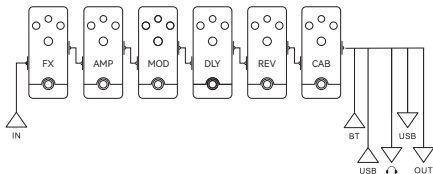
■ 放置场所:

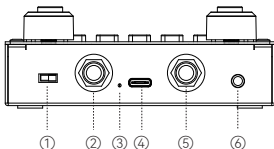
为防止变形、变色和其他方面等对产品造成损坏, 请避免以下情况:

- | | | |
|---------|------------|------------|
| a. 热源 | b. 磁场 | c. 阳光直射 |
| d. 高温潮湿 | e. 多尘或不洁地方 | f. 强烈震荡或摇晃 |

CN ANNBLACK BOX的效果模块连接图

(6个效果模块的顺序可自定义移动)





①电源开关。

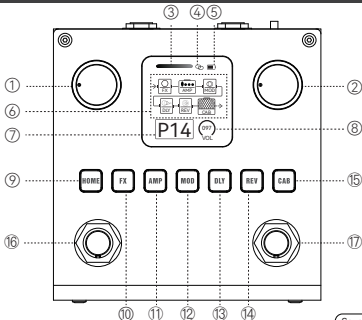
②IN：连接吉他/贝斯信号输入。

③充电指示灯：开机后，本机在正常使用时指示灯为熄灭状态。充电时指示灯为红灯常亮，充满电时灯会转色。

④USB接口：供电、电脑声卡、电脑数据交换传输、OTG内录或回放、外接MIDI踏板等功能。温馨提示:使用本机附带的USB线和另购的OTG转换器，可连接到手机第三方APP软件进行音画同步实时内录。

⑤OUT：连接音箱或调音台。

⑥耳机输出：连接监听耳机。



① **Master 旋钮**，控制音量大小（BT音量和USB输入音量不受此旋钮控制）。

② **PARA 旋钮**，带按键旋转编码器，旋转它可选择项目或者调节参数，单击它可进入、退出或确认项目，长按 1 秒可进入“Save to”预设保存界面，长按 2 秒可进入“Swap to”预设交换界面。在“Save to”或者“Swap to”界面时候，如果单击HOME键，可退出保存回到主界面。

Save to

<P08>

P08 Swap to

<P09>

③ **IN 输入孔信号 大小实时展示条**。

④ **无线 BT 图标**，图标熄灭时候代表 BT 未开启，白色图标表示 BT 已开启但是未连接，蓝色图标表示 BT 已成功连接。

⑤ **电池状态图标**。

⑥ **当前预设下，六个效果模块的连接顺序和开关状态**。图标熄灭代表效果未开启，图标点亮代表效果已开启，图标的状态对应着下面右边的六个按钮的状态。如果六个图标都是熄灭状态，则代表目前本机处于 Bypass 直通状态。

⑦ **预设编号。**利用 PARA 旋钮选中此编号后，旋转可快速切换预设。本机包含80个预设音色，默认前40个预设（P01-P40）含有出厂效果，其中第36-40个音色为贝斯音色。恢复出厂设置时可单独选择前40个预设数据恢复到出厂状态，也可一次恢复全部80个预设状态。

注意：进入手机 APP 或者电脑软件界面即可进行此操作。

⑧ **预设音量。**

⑨ **HOME 带灯按键/BT开关键(带BT状态灯)，**单击 HOME 键可在屏幕上显示“主界面”，在主界面时候再单击HOME 键可进入模块操作界面。长按HOME 键开启/关闭BT模块。按键灭灯时候代表 BT 未开启，按键灯闪烁时候表示 BT 已开启但是未连接，按键灯常亮时候表示 BT 已成功连接。

开机后显示的界面称为“主界面”，在这个界面内可以直观且快速的浏览到当前预设内重要信息的设定情况，还可以了解到电池情况和其他信息等等。



模块操作界面，在此界面下，可便于操作6个模块的连接顺序，也可利用两个踩钉自由选择某个模块的开关（Stomp mode单块模式）。



⑩ **FX 多功能模块带灯按键，**在任何时候，长按按键可以快速开启/关闭模块效果。单击此按钮可进入模块的参数菜单，再次单击可开启/关闭模块效果。当灯常亮时，表明此效果模块开启状态，灯熄灭即代表效果关闭。当屏幕上显示出参数菜单后，您可使用 PARA 编码器来调整屏幕上任意一个效果和参数，此时您也可以使用左踩钉脚控开/关当前模块的效果。

温馨提示：6个效果模块的按键都可支持上面这些功能。

⑪ **AMP 箱头前级模拟模块带灯按键。**

⑫ **MOD 周边效果模块带灯按键，**当屏幕上显示出 MOD 的参数菜单时候，如果利用右踩钉，还可以 Tap 打点获取效果的 Speed 速度参数。此时如果效果已经开启的话，按键的灯也会根据Speed参数的速度进行闪烁。

⑬ **DLY 延时效果模块带灯按键**，当屏幕上显示出 DLY 的参数菜单时候，如果利用右踩钉，还可以 Tap 打点获取效果的 Time 速度参数。此时如果效果已经开启的话，按键的灯也会根据Time 参数的速度进行闪烁。当屏幕处于 DLY 的参数菜单并且效果已经开启时，长踩右踩钉可以启动 hold 功能，在长踩 1.5 秒后机器会读取此刻的延时湿声，如果脚一直不松开，就会让这个hold 湿声一直保持。如果脚松开，就可以马上取消 hold 湿声（Freeze功能）。

⑭ **REV混响效果模块带灯按键**。

⑮ **IR CAB箱体模拟效果模块带灯按键**。

⑯ **左踩钉**，在主界面时，用于向下选择预设。在某些其他界面时，可用于开/关模块。当同时踩下左&右踩钉，进入调音模式，然后再单击任意一个踩钉即可退出调音。

⑰ **右踩钉**，在主界面时，用于向上选择预设。在模块操作界面时，可用于挑选模块；在 MOD 或者 DLY参数菜单时，可用于 Tap 打点获取速度；在 DLY 参数菜单并且效果已经开启时，长踩可用于开启 Hold 功能（延时湿音保持，Freeze声音冻结无限延音）。

请知晓：在某些界面时右踩钉不起任何作用。

温馨提示：在切换到调音界面，如图所示，在此界面有两个项目可供选择，左下角的Refer to是选择A音的参考频率430-450Hz，右下角的Mode是选择声音输出状态Bypass或者Mute。



一：FX多功能模块（FX模块可实现Noise Gate或NG+Boost或NG+Compress，三种搭配方式，可同时使用6+1效果。）

1. Noise Gate++ 降噪功能

- Gate参数：调节 AI 降噪的力度，调至0时为关闭降噪效果。

2. Boost激励功能（可叠加降噪功能）

- Gate参数：调节 AI 降噪的力度，调至0时为关闭降噪效果。
- Gain参数：调节激励效果的增益度。

3. Compress压缩功能（可叠加降噪功能）

- Gate参数：调节 AI 降噪的力度，调至0时为关闭降噪效果。
- Sustain参数：综合了压缩的阈值，压缩比率等等的参数，数值越大压缩效果越明显。
- Attack参数：信号超过阈值后压缩器的启动时间，数值越大，压缩启动时间越慢，听觉上的音头更加突出；数值越小，启动时间越快。
- Level参数：控制模块的最终输出音量。

4.AI Gate MS降噪功能，基于统计偏差跟进模型的智能算法。

- Gate：打开Gate的阈值。它会持续监测输入信号的电平强度。阈值设置得越高，Gate 打开的条件越苛刻，只有更强的信号才能通过；阈值越低，则更多信号可以通过。
- Bias：通过设置 Bias 提前量，可以让 AI 降噪模型提前“准备”好对即将出现的噪声或信号变化进行处理。在音频降噪中，如果能提前预测到噪声的变化趋势并相应地调整偏置，模型就能更准确地在噪声出现时进行抑制，减少噪声对有用信号的干扰，提高降噪的及时性和准确性。

二：AMP箱头前级模拟模块（1-15个为吉他音箱模拟，16-20个为贝斯音箱模拟）

- Gain
- Level
- Bass
- Mid
- Treble

三：Modulation周边效果模块（支持TAP功能）

1. Chorus合唱效果

- Speed：调节效果速度。
- Depth：调节效果深度。
- Mix：调节效果湿声和干声的比例。

2. Phaser相移效果

- Speed：调节效果速度。
- ParamA：MidCut，中频削减，数值越大，效果的中频削减越多。此参数主要为失真音色设计。
- ParamB：Resonance，相位效果的共振强度调节。
- Feedback：调节湿声的反馈量。

3. Tremolo颤音效果，使用低频振荡器来调制声音的输出电平。

- Speed：调节效果速度。
- Depth：调节效果深度。
- Level参数：控制模块的最终输出音量。

4. Flanger弗朗格效果，跟合唱类似的声音，但是使用一个更短的延时间并且增加反馈参数，产生一个强烈频率的扫描效果。

- Speed：调节效果速度。
- Depth：调节效果深度。
- Feedback：调节湿声里的扫频效果的反馈量。
- Mix：调节效果湿声和干声的比例。

5. Vibrato震音效果，信号的音调会有周期性的升高降低，其结果与声乐家使用的颤音技术相似。与合唱或 Flanger效果相反，没有直接信号与音高调制信号相结合。

- Speed：调节效果速度。
- Depth：调节效果深度。

6. Univibe多谐振荡器效果，一种独特的音色，合唱和相移两种效果的结合。

- Speed：调节效果速度。
- Depth：调节效果深度。
- Mix：调节效果湿声和干声的比例。

7. Autofilter自动滤波器，通过设定的频率使滤波器有规律地进行工作，吉他和贝斯都可以使用。

- Speed
- Min：滤波器扫频的频率最小值，数值开大后可让湿声的中低频部分更加突出。
- Max：滤波器扫频的频率最大值，数值开大后可让湿声的中高频部分更加突出。
- Mix
- Feedback

四：Delay延时效果模块（支持FREEZE和TAP功能）

1. Analog模拟延时效果，模拟电子管的信号的延迟效果，音色具有复古、温暖等特点。

- Time，延时效果时间速度参数，600-60bpm。
- Feedback，调节延迟的反馈次数。
- Mix，调节干湿比。
- Phaser，可在在Delay湿声增加相移效果，此参数可调节相移效果的使用量。
- Pitch，调节失谐效果的PITCH参数，相当于在延迟湿声加入了一点失谐效果，是一种轻微移调的声音，一种复古且迷人的声音。

2. Ducking闪避延迟，我们在处理延时湿音时候，在湿音发声前部分加入了降噪效果，从而抑制湿音的前部分声音，达到一种前面闪避的效果，有慢慢增大的声音听感。通过非常轻微的闪避，延迟的动态更多的是“感觉到”而不是“听到”。

- Time，延时效果时间速度参数，600-60bpm。
- Feedback，调节延迟的反馈次数。
- Mix，调节干湿比。
- Unpack 解除，参数调大后，可解除湿声后部分音量的突然上升感觉，从而达到一种更平缓的延时动态。
- Speed，可在在Delay湿声增加合唱效果，此参数可调节合唱效果的速度。
- Depth，合唱的深度。

3. dTape磁带延时，它复刻了磁带机特有的延时风格，你不仅能获得专业级别磁带机的温暖和丝滑，而且能模拟磁带在褶皱、偏差等等真实情况下的声音特效。

- Time，延时效果时间速度参数，600-60bpm。
- Feedback，调节延迟的反馈次数。
- Mix，调节干湿比。
- Grit，这个参数调整磁带机的偏差，从欠偏差到过偏差，听起来就像湿音过载的感觉。它用于调整延迟湿音的动态范围和余量。
- Speed，使用 speed 参数来控制磁带不规则的量和严重程度，包括摩擦、折痕、拼接和污染物。使用最小值以获得清新、干净的磁带延时，也可使用最大值来获得多年来被破坏和腐蚀的磁带声音。
- Depth，该参数改变与机械相关的磁带速度波动的量。这也导致自然磁带机产生带类似合唱的声音。在参数最小值时，可以获得更和谐的音色，类似一个状态很好的磁带机。参数最大值时，听到一个感觉需要维修的磁带机的声音。在中间的参数，实现了比较自然的磁带音色。

4. Dual双延迟，两个独立的延时回音，第一个回音和第二个回音的时差比例可达到非常有趣的延迟节奏效果。

- Time，延时效果时间速度参数，1200-120bpm。
- Feedback，调节延迟的反馈次数。
- Mix，调节干湿比。

- Time-Mode时间模式，调节两个回音之间的时差，最小值时候就相当于没有时差，数值越大时差越大。
- Speed，可在在Delay湿声增加合唱效果，此参数可调节合唱效果的速度。
- Depth，合唱的深度。

5. Lofi低保真延时，一种特殊、复古且具有破坏性的延时效果，延时湿音里体现了过滤器，黑胶唱片，噪音低保真等多种感觉。

- Time，延时效果时间速度参数，600-60bpm。
- Feedback，调节延迟的反馈次数。
- Mix，调节干湿比。
- Grit，当参数调大时，听起来像是湿音过载的感觉。
- Speed，可在在Delay湿声增加合唱效果，此参数可调节合唱效果的速度。
- Depth，合唱的深度。

五：Reverb混响效果模块

1. Room房间混响效果，房间混响模拟结构简单的房间声音，其中许多反射被房间中的软材料吸收，声音被墙壁反射。

- Decay，调整回声效果的持续时间。
- Mix，调节干声和混响湿音的音量比例。
- HPass，HIGH PASS高通，仅作用于湿音。
- LPass，LOW PASS低通，仅作用于湿音。
- Depth，使混响湿音产生音调的小幅度周期性升高降低，得到一种复古迷人的背景声音。

2. Hall大厅混响效果，大厅混响给人一种宽阔的，轻微散射的感觉。它能模拟宏大的环境声音。

- Decay
- Mix
- HPass
- LPass
- Depth

3. SWELL膨胀混响，在干信号之后逐渐带入混响效果，两个信号间进行自然的衔接，以获得轻微的上升效果和演变氛围感，就像有个可以控制混响湿音的音量踏板。

- Decay
- Mix，如果把干音去除，再把RiseT参数调到0，可以模仿提琴等弦乐乐器的音色。
- HPass
- LPass
- RiseTime上升时间，当参数加大时，上升时间会变长，听觉上的膨胀效果会减弱，适合用于演奏和弦节奏；反之上升时间变短时候，适合用于演奏旋律。

4. Spring弹簧混响效果，弹簧混响是常见的混响类型，声音信号放大到弹簧上，然后拾音器捕捉到弹簧的共振声，从而形成人们在空间中听到的混响，弹簧混响的特点是来自弹簧的独特颤抖感。

- Decay
- Mix
- HPass
- LPass
- Combs，控制弹簧的数量。

5. Shimmer微光混响效果，微光混响，是一种在混响湿音中存在音高偏移的声音。通过PITCH参数的调节，可调整微光声音的音高，可调出不和谐音程，营造出恐怖的背景声音，这是非常创造性的用法。也可调出和谐音程，创造出一种自带美颜的声音，在混响尾音部分会听到犹如微光汲汲传动的美妙声音，如一种天堂中的声音，给人感觉像在开阔的土壤上，遇见日出从微光到全面灿烂的过程。

- Decay
- Mix
- Tone，调节混响湿声的明亮度。
- Pitch，调节微光声音的音高，最小值为音高跟原声一致，最大值为比原声高两个八度。
- Amount，控制微光声音的用量。

6. Cloud云混响效果，一种华丽的大型环境混响，听起来像是声音从云层卷积而来，仅仅弹一个音符，在混响湿音中就出现非常多的声音从各处传来，声音听起来非常宽广。

- Decay
- Mix
- HPass
- LPass
- Diffusion扩散度，可调节环境维度感，当参数增大时，早期反射的混响湿音被软化，听起来更平滑，模糊了延迟和混响之间的界限。

六：IR CAB箱体模块（1-15为电吉他箱体模拟，16-20为贝斯箱体模拟）

- Level，箱体输出音量。
- Low cut，调节低切频点，可选择20-300赫兹。如果您选择 300 赫兹，意味着 300 赫兹以下的低频都被抑制。
- High cut，调节高切频点，可选择18000-5000赫兹。如果您选择 5k 赫兹，意味着 5000 赫兹以上的高频都被抑制。

CN 技术参数

输入	标准1/4" 单声道接口/TS
输出	标准1/4" 单声道接口/TS, 1/8"立体声输出
采样率	44.1KHz/24bit
尺寸	113(L)x109(W)x41(H)mm
重量	410g
电源供给& USB数据传输	Type-C、5V≥300mA
电池额定容量	2000mah
电池型号规格	103450
电池标称电压	3.7V
续航	10+ 小时
充电电流限制	约408毫安
动态范围	105dB
总效果数量	60+ 个
效果槽/效果模块	可同时使用 6+1个, 可自定义效果顺序
预设数量	40+40个
USB音频(OTG内录)	44.1K/24Bit, 内录 Loopback 可以独立开关
USB MIDI	支持
BT伴奏	支持
音色编辑软件支持	WINDOWS,MAC,IOS,Android
第三方IR文件格式 要求	44.1K/24Bit, 单声道 WAV 文件, 1024采样点
供电方式	内置蓄电池或USB 5V
AMP 音箱模拟	20种 (15 个电吉他音箱& 5 个贝斯音箱) 支持 加载第三方AMP 音色
IR CAB 箱体模拟	20种 (15 个电吉他箱体& 5 个电贝斯箱体) 支 持加载第三方 IR 文件
工作电流	约200mA
出厂附带	USB连接线X1
修订	2025-06-16

● **手机 APP 连接步骤：**在操作前需确认手机和 ANNBLACK BOX 的 BT 功能都已经开启，然后打开 M-EFCS，点击左上角“≡”按键，点击“当前设备”选择 ANNBLACK BOX，再扫描附近设备，即可连接使用。



扫描二维码下载

● **可支持有线/无线连接 MIDI 踏板**（MIDI 踏板需另购，以 Chocolate Plus 为例），从而达到拓展踩钉的目的。实现以下功能：可实现 4 个预设为一个 Bank，可 Bank+ 和 Bank-，可自由跳跃预设音色。可通过踩钉，一键开关 6 个模块。可使用第一个踩钉代替 Home 键，第二个踩钉代替 PARA 按键的“按下”的动作，第三和第四个踩钉代替 PARA 按键的“逆时针转”和“顺时针转”。

● **温馨提示：**Loopback 功能，是常见的声卡回环功能，可到软件上开启或者关闭，默认出厂为开启状态。功能开启时，电脑/手机内的音乐声会与乐器声同时录制，同一轨道混合录制，其中包括 USB 和 BT 的音乐声。如果功能关闭，使用 Blackbox 作为声卡进行录音时候，只能录制到 IN 接口的乐器声，这样更便于电脑录音时候进行乐器声分轨录音。

● 我们将不断改进我们的产品并保留修改某些功能和规格(包括但不限于外观、包装设计、操作手册、配件、尺寸、规格参数、显示屏等)的权利,恕不另行通知。购买前请与当地经销商确认产品功能及规格。产品图片及颜色因拍照光线误差及屏幕设定可能与实物产品效果有所差异。请以实物为准。